1 Autonomous Energy Production for Isolated Site (CORRECTION)

Prelim

1. $$E_{\text{year}} = \overbrace{300 \times 24 \times 365}^{\text{average}} + \overbrace{1500 \times 3 \times 0.5 \times 365}^{\text{peak}} = 3.45 \text{ MWh/year}$$
$$= 9.45 \text{ kWh/day}$$

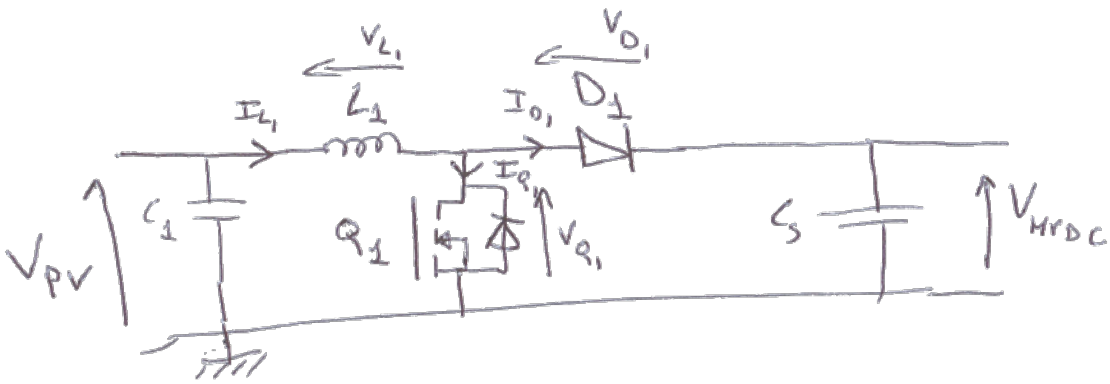
2. With 1520 kWh/m^2/year

$$\left.\begin{array}{l}\eta_{\text{PV}} = 20\% \\ \eta_{\text{Elec}} = 90\%\end{array}\right\} 18\% \Rightarrow 270 \text{ kWh/m}^2\text{/year (Elec energy)}$$

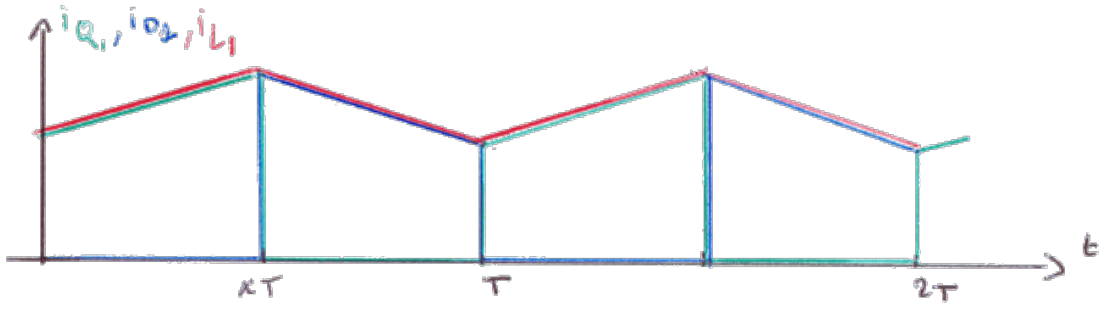
$$S = \frac{3.45 \cdot 10^3}{270} = 12.8\text{m}^2$$

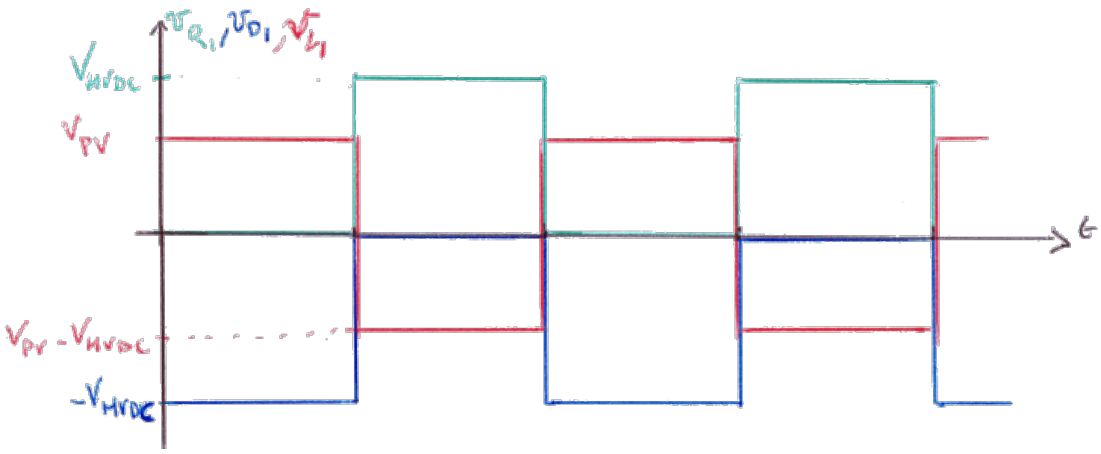
3. Number of panels: $\frac{12.8}{1{,}13 \times 1{,}06} = 10,7 \Rightarrow 11$ panels. ($1.13 =$ Surface of 1 panel)

1.1 Boost Converter



1.





2.

$$< v_{L1} >= 0 \Rightarrow \alpha V_{PV} + (1-\alpha)(V_{PV} - V_{HVDC}) = 0$$

$$\Rightarrow V_{HVDC} = \frac{V_{PV}}{1-\alpha}$$

- <u>No losses</u> $\Rightarrow V_{PV} \cdot I_{PV} = V_{HVDC} \cdot I_{HVDC}$

  $\Rightarrow IPV = \dfrac{I_{HVDC}}{1-\alpha}$

3.

	Max Voltage	Max Current	Average Current
L_1	$Max(V_{PV}; \lvert V_{PV} - V_{HVDC}\rvert)$	$I_{PV} + \frac{\Delta I}{2}$	I_{PV}
Q_1	V_{HVDC}	$I_{PV} + \frac{\Delta I}{2}$	αI_{PV}
D_1	V_{HVDC}	$I_{PV} + \frac{\Delta I}{2}$	$(1-\alpha) I_{PV}$

4.
   - STC : 1 pannel : $250\text{W}_C \Rightarrow$ 11 panels : $P_{TC} = 3000\text{W}_C$
   - Through each BOOST : $\dfrac{3000}{2} = 1500\text{W}$
   - In the battery : $3000 - \underbrace{1500 - 300}_{\text{max power consumed}} = 1200$ W

5. 5.1 1 panel @ STC $\rightarrow V_{1PV} = 39V \Rightarrow$ 6 panels in series $\rightarrow V_{PV} = 234$ V

   5.2 1 Boost $\rightarrow P = 1500$ W $\Rightarrow I_{PV} = \dfrac{1500}{234} = 6.4$ A

   5.3 $V_{HVDC} = 500$ V $\Rightarrow \alpha = \dfrac{V_{HVDC} - V_{PV}}{V_{HVDC}} = 0.53$

   5.4 $I_{PV} = 6.4$ A $\Rightarrow \Delta I = 1.28$ A (20% if I_{PV})

   $\underline{0 < t < \alpha t}$: $V_L = V_{PV} = L\dfrac{i_{PV}}{dt} \Rightarrow \Delta I = \dfrac{V_{PV}}{L}\alpha T$

   $\Rightarrow L = 1.93$ mH

   5.5

	Max Voltage	Max Current	Average Current
Q_1	500 V	7 A	3.4 A
D_1	500 V	7.1 A	3 A

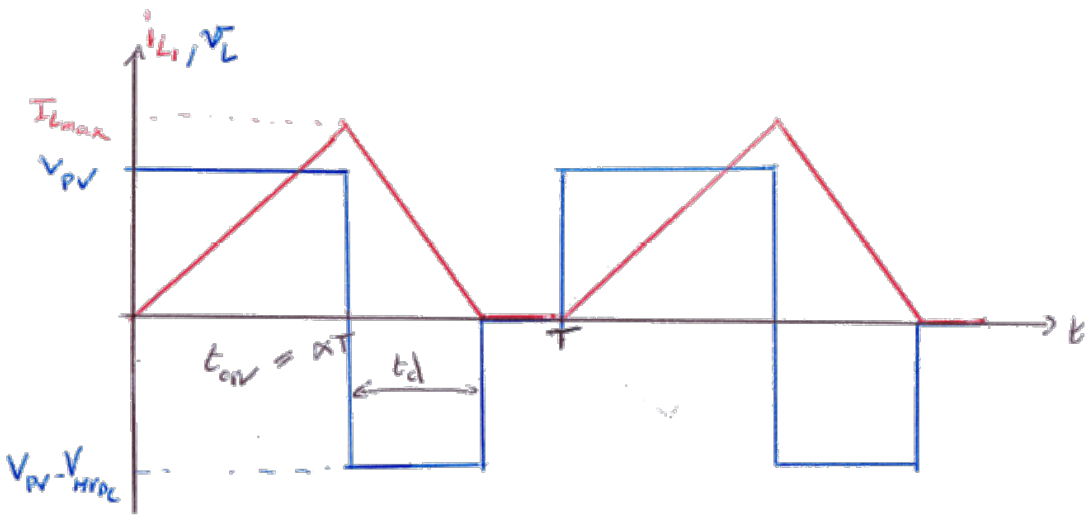
6. 6.1 $V_{PV} = 6 \times 44 = 264$ V

   $\begin{cases} \text{Battery charged} & : P_{batt} = 0 \text{ W} \\ \text{min load} & : P_{load} = 300 \text{ W} \end{cases} \rightarrow$ 1 BOOST : $P = 150$ W

   $I_{PV} = \dfrac{150}{264} = 0.57$ A

   6.2 $I_{PV} < \dfrac{\Delta I}{2} \Rightarrow$ <u>Discontinuous mode</u>

   6.3



6.4 $< v_L >= 0 \Rightarrow V_{PV}\ cdot I_{PV} = \frac{I_{Lmax}}{2} \cdot \frac{t_{on} + t_D}{T}$

$\underline{0 < t < \alpha T}:\ v_L = V_{PV} = L\frac{di_L}{dt} \Rightarrow I_{Lmax} = \frac{V_{PV}}{L} t_{on}$

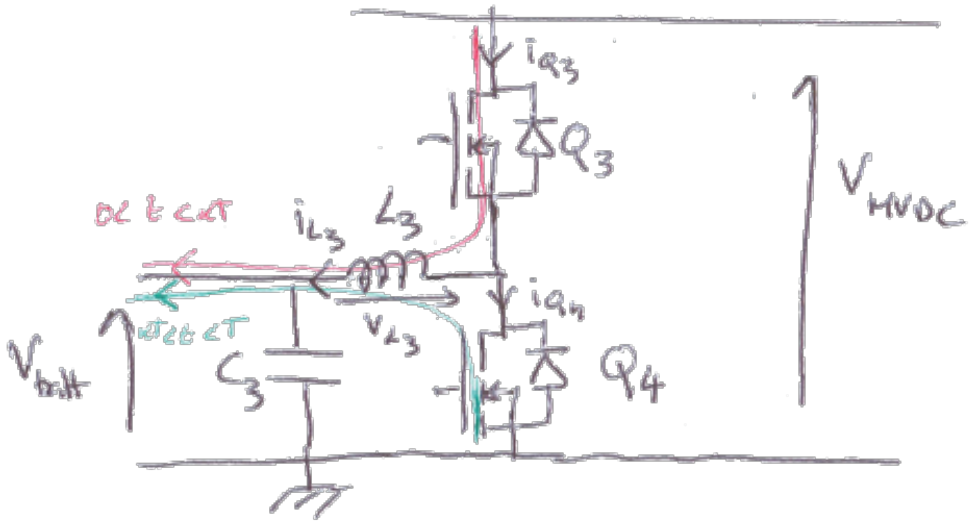
$\Rightarrow I_{PV} = \frac{V_{PV}}{2L} t_{on} \cdot F_d \cdot t_{on} \left(1 + \frac{V_{PV}}{V_{HVDC} - V_{PV}}\right)$

$\Rightarrow I_{PV} = \frac{V_{PV} V_{HVDC}}{V_{HVDC} - V_{PV}} \cdot \frac{F_d}{2L} \cdot t_{on}^2$

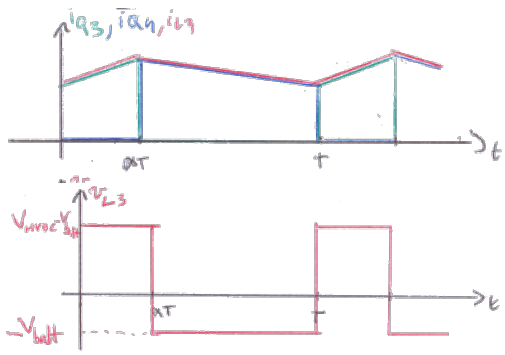
6.5 $t_{on} = 8.8\mu\text{s} \rightarrow \alpha = 0.44$

1.2 Battery Charger

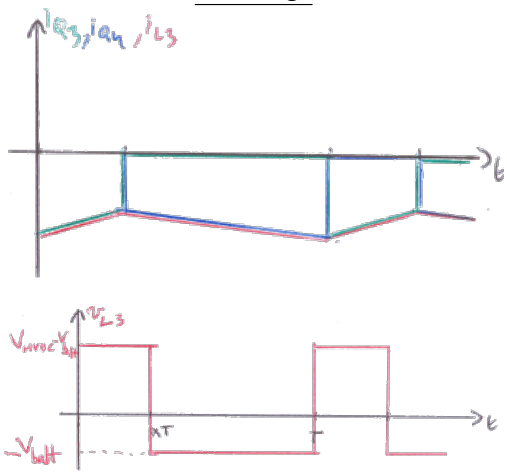
1.



Charge



Discharge



2. $< v_L >= 0 \Rightarrow V_{batt} = \alpha_{HB} V_{HVDC}$

3.
   - $P_{batt} = 1200$ W $\rightarrow$ battery charging $\rightarrow$ Half Bridge in Buck
   - C_3 is used to filter hight frequency current ripple so it does not flow in the Battery

4. 4.1 $I_{batt} = \dfrac{P_{batt}}{V_{batt}} = \dfrac{1200}{120} = 10$ A $\rightarrow \Delta I = 3$ A

   $V_{batt} = 120$ V, $V_{HVDC} = 500$ V $\rightarrow \alpha_{HB} = 0.24$

4.2 $\underline{0 < t < \alpha T}$: $V_L = V_{HVDC} - V_{batt} = L\dfrac{di_L}{dt}$

$$\Rightarrow \Delta I = \frac{V_{HVDC} - V_{batt}}{L}\alpha T \Rightarrow L = 600\mu\text{H}$$

5.

	Max Voltage	Max Current	Average Current
Q_3	500 V	11.5 A	2.4 A (αI_{batt})
Q_4	500 V	11.5 A	7.6 A ($(1-\alpha)I_{batt}$)

$$(I_{bat} + \frac{\Delta I}{2})$$

6. $\begin{cases} P_{load} = 1800 \text{ W} \\ P_{PV} = 1000 \text{ W} \end{cases} \rightarrow \begin{cases} P_{batt} = -800 \text{ W} \\ I_{batt} = -6.7 \text{ A} \end{cases}$

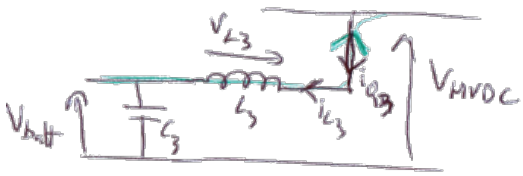
7. Half Bridge in Boost operation $\rightarrow$ energy supplied by the Battery

8. $\alpha_{HB} = 0.24$ (V_{HVDC} and V_{bat} are still the same)

   No discontinuous mode for Half Bridge (current reversible)

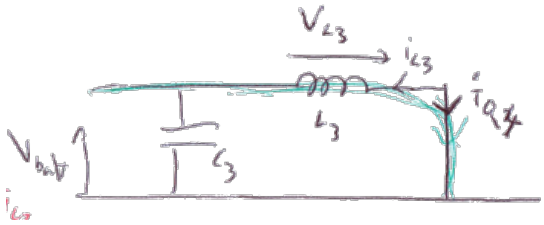
   $\rightarrow$ current in L_3 is just negative

$\underline{0 < t < \alpha_{HB}T}$:

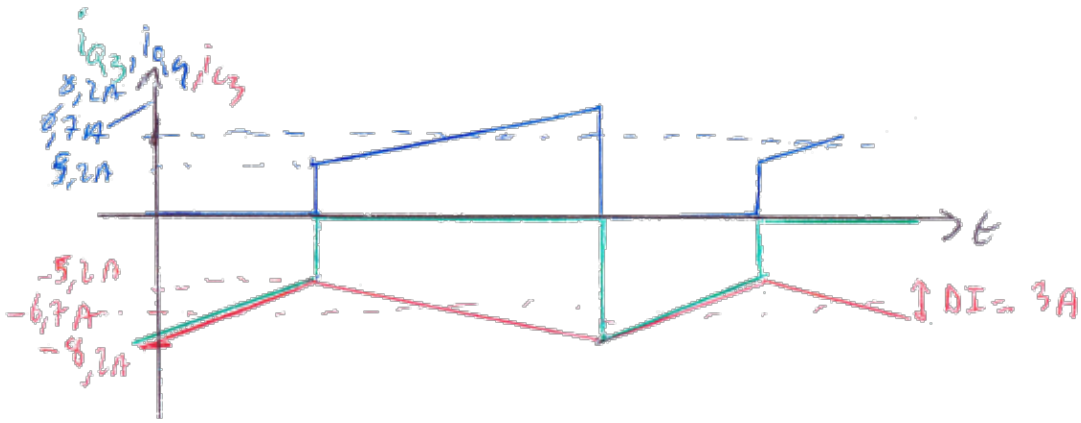


$V_{L_3} = V_{HVDC} - V_{batt}$

$\underline{\alpha_{HB}T < t < T}$:

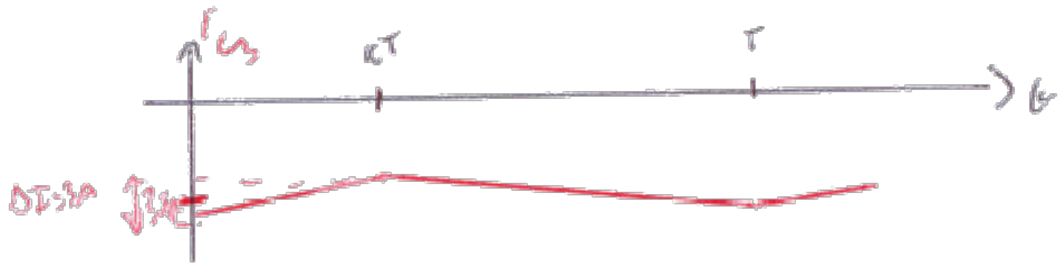


$V_{L_3} = -V_{batt}$



9. 9.1 $P = 300$ W $\rightarrow I_{batt} = -2.8$ A

   9.2 $\alpha_{HB} = 0.24$ (Still same values of V_{batt} and V_{HVDC})



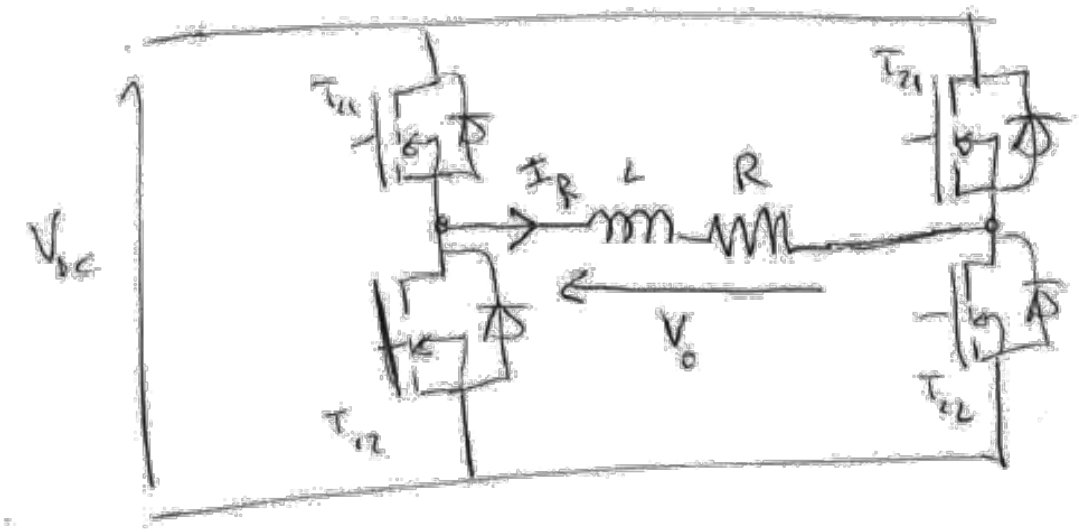
10. $P = 300$ W $\qquad \eta = 90\% \Rightarrow P_{batt} = 333$ W

    For 16h $\rightarrow E_{batt} = 5.3$ kWh

    Max discharge 90% $\rightarrow E_{battTOTAL} = \dfrac{E_{batt}}{0.9} = 5.9$ kWh

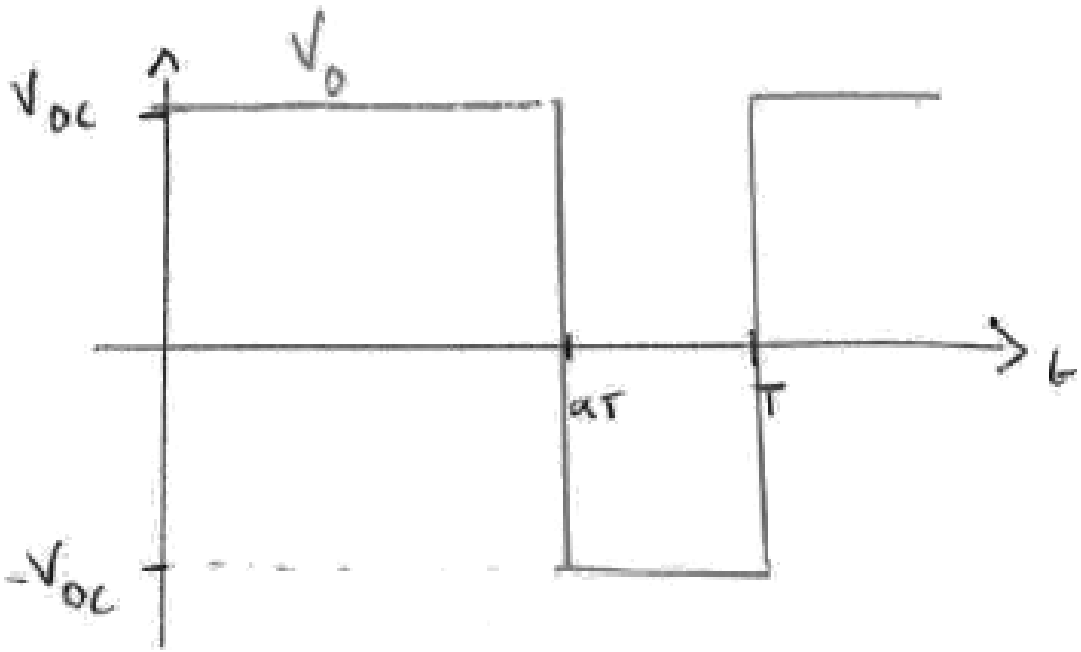
    $V_{batt} = 120$ V $\Rightarrow C_{batt} = 50$ Ah

1.3 Single-Phase PWM Inverter on RL Load



$$\alpha(t) = 0.5 + \delta_\alpha \sin(\omega t)$$

1. 1.1



1.2 $< V_0 >= (2\alpha - 1)V_{DC}$

1.3 $< V_0 > (t) = (2\alpha(t) - 1)V_{DC} = 2\delta_\alpha V_{DC} \sin(\omega t)$

1.4 $V_{0eff} = \sqrt{2}\delta_\alpha V_{DC}$

1.5 $I_{Reff} = \dfrac{V_{0eff}}{\sqrt{R^2 + (L\omega)^2}}$

1.6 $\tan\delta = \dfrac{L\omega}{R} \Rightarrow \varphi = \arctan(\dfrac{L\omega}{R})$

1.7 $pf = \cos\varphi = cos(\arctan(\dfrac{L\omega}{R}))$

1.8

$$P_R = R \cdot I_{Reff}^2 = \frac{RV_{0eff}^2}{R^2 + (L\omega)^2} = \frac{2R\delta_\alpha^2 V_{DC}^2}{R^2 + (L\omega)^2}$$
$$= \frac{2\delta_\alpha^2 V_{DC}^2}{R} \cdot \frac{1}{1 + \tan^2\varphi}$$

2. 2.1 $P_R = 1500$ W $\qquad U_{Rn} = 230$ V $\Rightarrow I_{Rn} = \dfrac{P_R}{U_{Rn}} = 6.5$ A

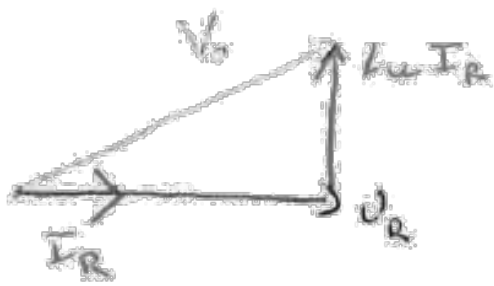
$P_R = R \cdot I_{Rn}^2 \Rightarrow R = \dfrac{P_R}{I_{Rn}^2} = 35.3\Omega$

Or $P_R = 500$ W $= \dfrac{U_{Rn}^2}{R} \Rightarrow R\dfrac{U_{Rn}^2}{P_R} = \dfrac{230^2}{1500} = 35.3\Omega$

2.2 $L = 30$ mH $\qquad f = 50$ Hz $\Rightarrow \varphi = \arctan(\dfrac{L\omega}{R}) = 0.26(15°)$

$\cos\varphi = 0.97$

2.3



2.4 $I_R = 6.5$ A $\rightarrow V_L = L\omega I_R = 61.23$ V

2.5 $V_{0eff} = \sqrt{U_R^2 + V_L^2} = 238$ V

2.6 $V_{0eff} = \sqrt{2}\delta_\alpha V_{DC}$ with $V_{DC} = 500$ V $\Rightarrow \delta_\alpha = \dfrac{V_{0eff}}{\sqrt{2}V_{DC}} = 0.34$

3. Max voltage : $\dfrac{V_{DC}}{2} = 250$ V

   Max current : $\dfrac{V_{0eff}\sqrt{2}}{\sqrt{R^2 + (L\omega)^2}} = \dfrac{\sqrt{2}\delta_\alpha V_{DC}\sqrt{2}}{\sqrt{R^2 + (L\omega)^2}} = 9.3$ A